

Operators Without Nontrivial Invariant Subspaces on $C_b(K)$ A Complete Classification and Literature Audit

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Abstract

We audit a proposed argument removing completeness from the final $C(K)$ theorem of Cheng–Fang–Jiang. The argument is correct: for every metric space K with infinite-dimensional $C_b(K)$, the space $C_b(K)$ admits a bounded operator without nontrivial closed invariant subspaces if and only if K is compact. In fact, this follows from classical ingredients: Conway’s characterization of separable $C_b(K)$ for completely regular spaces, Sobczyk’s complementation theorem for c_0 , and Read’s construction of operators without invariant subspaces.

We give a self-contained proof and a complete all-dimensional formulation. For a nonempty topological space K , let K_{cr} be its bounded-continuous-function reflection. Then $C_b(K)$ supports an operator with no nontrivial closed invariant subspaces if and only if K_{cr} is compact metrizable and is either a singleton or an infinite set. Under the source paper’s infinite-dimensional hypothesis, this is equivalent to separability of $C_b(K)$. For Tychonoff spaces, and hence for metric spaces, the reflection is K itself. This result is separate from Halmos’s third problem, which it does not resolve.

1 Context and verdict

Let K be a topological space and let $C_b(K)$ denote the Banach space of bounded continuous complex-valued functions on K , with the supremum norm. For a Banach space X and $T \in \mathcal{B}(X)$, write

$$\text{Lat}(T) = \{M \subseteq X : M \text{ is closed and linear, and } T(M) \subseteq M\}.$$

We call T *transitive* when $\text{Lat}(T) = \{0, X\}$.

Cheng–Fang–Jiang proved that, for a complete metric space K with infinite-dimensional $C_b(K)$, such a transitive operator exists on $C_b(K)$ if and only if K is compact [1, Theorem 16.1]. The proposed partial argument removes completeness by combining a closed discrete sequence with Tietze extension.

Verdict. The partial argument is mathematically correct. The only point worth making explicit is that Tietze extension is first applied to the real and imaginary parts of a complex-valued function. More importantly, the exact metric statement is not an unresolved problem: it follows immediately from a classical theorem of Conway that $C_b(K)$ is separable exactly when a Tychonoff space K is compact metrizable [2, Theorem V.6.6, p. 140], together with Read’s and Sobczyk’s theorems. Sections 2–5 give a complete proof and a stronger classification.

16. PROOF OF THEOREM 1.9

Based on another beautiful result of Read [45] that an infinite dimensional separable complex Banach space X of the form $X = c_0 \oplus Y$ for some separable Y always admits an operator $T \in \mathfrak{B}(X)$ without any non-trivial invariant subspace, and the Sobczyk theorem [48] that c_0 is 2-complemented in every separable Banach space geometrically containing it, in this section, we show the following theorem.

Theorem 16.1. *Let K be a complete metric space such that the space $C(K)$ of all bounded continuous complex valued functions is infinite dimensional. Then a sufficient and necessary condition for $C(K)$ admitting an operator $T \in \mathfrak{B}(C(K))$ without any nontrivial subspace is that the metric space K is compact.*

Figure 1: The source theorem assumes that the metric space K is complete.

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Then it is easy to observe that $C_0(K)$ is a closed subspace of $C(K)$ isometric to c_0 . Therefore, again by the Sobczyk theorem [48], c_0 is 2-complemented in $C(K)$. Therefore, the sufficiency follows from the previously mentioned theorem of Read [45].

Necessity. Note that every bounded linear operator on a nonseparable Banach space admits a nontrivial invariant subspace. Since K is a complete metric space, and since that there is $T \in \mathfrak{B}(C(K))$ without any nontrivial subspace, $C(K)$ must be separable and infinite dimensional. Therefore, the cardinality of K is infinite. If K is not compact, then it is not difficult to observe that ℓ_∞ can be linearly isometrically embedded into $C(K)$. This contradicts to that $C(K)$ is separable. □

Figure 2: The proof tail where completeness enters the source argument.

2 Audit of the proposed metric argument

The two lemmas in the proposed solution are both valid.

Proposition 2.1. *Let X be a nonseparable Banach space. Then every $T \in \mathfrak{B}(X)$ has a nontrivial closed invariant subspace.*

Proof. Choose $0 \neq x \in X$ and set

$$M_x = \overline{\text{span}}\{x, Tx, T^2x, \dots\}.$$

Then M_x is nonzero, closed, and T -invariant. It is separable because it is generated by a countable set. Since X is nonseparable, $M_x \neq X$. □

The same observation gives a useful converse formulation.

Corollary 2.2. *If a Banach space supports a transitive bounded operator, then it is separable. Indeed, every nonzero vector is cyclic for that operator.*

Lemma 2.3. *Let X be a nonzero finite-dimensional complex Banach space. A transitive operator exists on X if and only if $\dim X = 1$.*

Proof. If $\dim X = 1$, there are no nonzero proper linear subspaces. If $\dim X \geq 2$, every $T \in \mathcal{B}(X)$ has an eigenvalue over $\mathbb{C}$. The corresponding eigenline is a nonzero proper closed invariant subspace. $\square$

Proposition 2.4. *If K is a noncompact metric space, then $C_b(K)$ is nonseparable.*

Proof. Metric compactness is equivalent to sequential compactness. Hence there is a sequence in K with no convergent subsequence. After passing to a subsequence, assume its terms x_n are distinct and put

$$D = \{x_n : n \in \mathbb{N}\}.$$

The set D is closed and discrete in K : an accumulation point of D would produce a convergent subsequence.

Every bounded function on D is continuous, so $C_b(D)$ is isometric to ℓ_∞ . The restriction map

$$R : C_b(K) \longrightarrow C_b(D), \quad Rf = f|_D,$$

is bounded and linear. It is surjective. Indeed, given $a : D \rightarrow \mathbb{C}$ bounded, apply the real-valued Tietze extension theorem to $\operatorname{Re} a$ and $\operatorname{Im} a$ separately; metric spaces are normal and D is closed. Combining the two extensions gives a bounded continuous complex-valued extension of a to K .

Thus ℓ_∞ is a continuous linear image of $C_b(K)$. Since a continuous image of a separable metric space is separable, while ℓ_∞ is nonseparable, $C_b(K)$ cannot be separable. $\square$

Corollary 2.5. *Let K be metric. If $C_b(K)$ supports a transitive bounded operator, then K is compact.*

Proof. By Corollary 2.2, $C_b(K)$ is separable. Proposition 2.4 excludes noncompact K . $\square$

This verifies the new part of the proposed proof. Notice that completeness plays no role.

3 The classical separability theorem for $C_b(K)$

A space is called *Tychonoff* if it is completely regular and Hausdorff. The following theorem is classical; the proof is included to make the logical closure of the argument transparent.

Theorem 3.1 (Conway). *Let K be a Tychonoff space. Then $C_b(K)$ is separable if and only if K is compact metrizable.*

Proof. Suppose first that $C_b(K)$ is separable, and choose a norm-dense sequence $(f_n)_{n \geq 1}$ in $C_b(K)$. Define

$$\Phi : K \longrightarrow \mathbb{C}^{\mathbb{N}}, \quad \Phi(x) = (f_1(x), f_2(x), \dots),$$

where $\mathbb{C}^{\mathbb{N}}$ has the product topology.

We claim that Φ is a topological embedding. If $x \neq y$, complete regularity gives $g \in C_b(K)$ with $g(x) = 0$ and $g(y) = 1$. Choose n with $\|f_n - g\|_\infty < 1/3$. Then $f_n(x) \neq f_n(y)$, so Φ is injective.

To see that the inverse on $\Phi(K)$ is continuous, let $x \in U$ with U open in K . Complete regularity gives $g \in C_b(K)$ such that $g(x) = 0$ and $g = 1$ on $K \setminus U$. Choose n with $\|f_n - g\|_\infty < 1/4$. Then

$$V = \{y \in K : |f_n(y) - f_n(x)| < 1/2\}$$

is an open neighborhood of x contained in U . Hence the coordinates f_n generate the topology of K , proving the claim. Therefore K is metrizable. If it were noncompact, Proposition 2.4 would imply that $C_b(K)$ is nonseparable, a contradiction. Thus K is compact metrizable.

Conversely, suppose K is compact metrizable. Choose a countable dense set $\{x_j : j \geq 1\}$ in K . The unital algebra over $\mathbb{Q} + i\mathbb{Q}$ generated by the real-valued functions $d(\cdot, x_j)$ is countable, self-adjoint, and separates points. The complex Stone–Weierstrass theorem therefore makes it uniformly dense in $C(K) = C_b(K)$. Thus $C_b(K)$ is separable. $\square$

For metric spaces, Theorem 3.1 directly says that the completeness hypothesis in [1, Theorem 16.1] is unnecessary. The Tietze proof in Section 2 is an elementary proof of the only nontrivial direction needed for that corollary.

4 The compact-metric existence direction

For completeness, we now supply the operator-theoretic direction rather than merely invoking Cheng–Fang–Jiang.

Proposition 4.1. *Let L be an infinite compact metric space. Then $C(L)$ admits a bounded operator with no nontrivial closed invariant subspaces.*

Proof. First we exhibit an isometric copy of c_0 in $C(L)$. Since L is infinite and compact metric, it contains a sequence of distinct points converging to a point $p \in L$. Passing to a subsequence, write the points as x_n and set $r_n = d(x_n, p) > 0$ so that

$$r_{n+1} < \frac{1}{4}r_n \quad (n \geq 1).$$

Define

$$u_n(x) = \max \left\{ 0, 1 - \frac{4d(x, x_n)}{r_n} \right\}, \quad x \in L.$$

Then $0 \leq u_n \leq 1$, $u_n(x_n) = 1$, and the supports of the u_n are pairwise disjoint. Indeed, for $m > n$,

$$d(x_n, x_m) \geq r_n - r_m > \frac{3}{4}r_n,$$

whereas points in both supports would force $d(x_n, x_m) \leq r_n/4 + r_m/4 < 5r_n/16$.

For $a = (a_n) \in c_0$, set

$$J(a) = \sum_{n=1}^{\infty} a_n u_n.$$

The series converges uniformly because the supports are disjoint and $a_n \rightarrow 0$. Moreover

$$\|J(a)\|_{\infty} = \sup_n |a_n|,$$

so J is an isometric embedding of c_0 into $C(L)$.

The space $C(L)$ is separable. Sobczyk’s theorem says that every copy of c_0 in a separable Banach space is complemented [5]. Hence

$$C(L) \cong c_0 \oplus Y$$

for some separable Banach space Y . Read proved that every separable complex Banach space of the form $c_0 \oplus Y$ admits a bounded operator without nontrivial closed invariant subspaces [4]. Transporting that operator through the above isomorphism gives the required operator on $C(L)$. $\square$

Remark 4.2. The complemented form of Read's theorem used here is exactly the Read–Sobczyk route invoked by Cheng–Fang–Jiang. No explicit formula for Read's operator is needed for the classification.

5 A full classification for arbitrary topological spaces

The metric theorem has a natural maximal extension. For an arbitrary topological space K , define the evaluation map

$$e_K : K \longrightarrow \mathbb{C}^{C_b(K)}, \quad e_K(x) = (f(x))_{f \in C_b(K)},$$

and let

$$K_{\text{cr}} = e_K(K)$$

with the subspace topology inherited from the product. The space K_{cr} is Tychonoff. It is the reflection of K detected by its bounded continuous scalar-valued functions.

Lemma 5.1. *Pullback along e_K is an isometric isomorphism*

$$U_K : C_b(K_{\text{cr}}) \longrightarrow C_b(K), \quad U_K g = g \circ e_K.$$

Proof. Because e_K is onto K_{cr} , pullback is isometric and injective. For each $f \in C_b(K)$, the corresponding coordinate projection

$$\hat{f} : K_{\text{cr}} \rightarrow \mathbb{C}, \quad \hat{f}((z_h)_{h \in C_b(K)}) = z_f,$$

is bounded and continuous, and $\hat{f} \circ e_K = f$. Thus U_K is surjective. $\square$

Theorem 5.2 (Complete classification). *Let K be a nonempty topological space. There is an operator $T \in \mathcal{B}(C_b(K))$ with $\text{Lat}(T) = \{0, C_b(K)\}$ if and only if the bounded-function reflection K_{cr} is compact metrizable and K_{cr} is either a singleton or infinite.*

Proof. Suppose first that such an operator exists. Corollary 2.2 shows that $C_b(K)$ is separable. By Lemma 5.1, $C_b(K_{\text{cr}})$ is separable. Since K_{cr} is Tychonoff, Theorem 3.1 implies that K_{cr} is compact metrizable. If K_{cr} is finite of cardinality m , then

$$C_b(K) \cong C(K_{\text{cr}}) \cong \mathbb{C}^m.$$

Lemma 2.3 forces $m = 1$.

Conversely, if K_{cr} is a singleton, then $C_b(K) \cong \mathbb{C}$, so there is no nonzero proper subspace to be invariant. If K_{cr} is infinite compact metrizable, Lemma 5.1 identifies $C_b(K)$ isometrically with $C(K_{\text{cr}})$, and Proposition 4.1 supplies a transitive operator on the latter. Conjugating by the isometric isomorphism gives one on $C_b(K)$. $\square$

Corollary 5.3 (Infinite-dimensional form). *Let K be a topological space and suppose $C_b(K)$ is infinite-dimensional. The following are equivalent:*

- (i) $C_b(K)$ supports a bounded operator without nontrivial closed invariant subspaces;
- (ii) $C_b(K)$ is separable;
- (iii) K_{cr} is compact metrizable.

Proof. The equivalence of (ii) and (iii) follows from Lemma 5.1 and Theorem 3.1. Since $C_b(K)$ is infinite-dimensional, K_{cr} is infinite, so Theorem 5.2 gives (i) $\Leftrightarrow$ (iii). $\square$

Corollary 5.4 (Tychonoff version). *Let K be a nonempty Tychonoff space. Then $C_b(K)$ admits a bounded operator without nontrivial closed invariant subspaces if and only if K is compact metrizable and K is either a singleton or infinite.*

Proof. For Tychonoff K , the evaluation map e_K is a homeomorphism from K onto K_{cr} . Apply Theorem 5.2. $\square$

Corollary 5.5 (Metric version, all dimensions). *Let K be a nonempty metric space. Then $C_b(K)$ admits a bounded operator without nontrivial closed invariant subspaces if and only if K is compact and K is either a singleton or infinite.*

Corollary 5.6 (Metric version under the source hypothesis). *Let K be an arbitrary metric space and suppose $C_b(K)$ is infinite-dimensional. Then $C_b(K)$ admits a bounded operator without nontrivial closed invariant subspaces if and only if K is compact.*

This is precisely the desired removal of completeness.

Remark 5.7 (Why a separation hypothesis matters). The conclusion “ K must be compact” cannot be extended verbatim to all topological spaces. Let P be an infinite set with the particular-point topology: fix $p \in P$ and declare every nonempty open set to contain p . Then P is noncompact, but every continuous map from P to a Hausdorff space is constant. If

$$K = [0, 1] \sqcup P$$

is the topological disjoint union, then K is noncompact while

$$C_b(K) \cong C([0, 1]) \oplus_{\infty} \mathbb{C} \cong C([0, 1] \sqcup \{*\}).$$

The latter is a separable infinite-dimensional $C(L)$ space with L compact metric, so it supports a transitive operator. The correct arbitrary-space criterion is therefore Theorem 5.2, phrased in terms of K_{cr} .

6 Literature outcome and scope

The literature search leads to the following conclusions.

1. The proposed Tietze argument is correct and supplies an elementary proof for arbitrary metric spaces.
2. A full answer was already implicit in classical literature. Conway’s theorem identifies separability of $C_b(K)$ with compact metrizability for Tychonoff K ; Read and Sobczyk supply the transitive operator on the compact-metric side. Thus there is no counterexample to the proposed metric statement.
3. The reflection theorem above packages the same ingredients for arbitrary topological spaces and also records the finite-dimensional exceptions. We have not located an earlier source that states this exact packaging, but its ingredients are classical, so no novelty claim should be made without a more exhaustive bibliographic investigation.
4. None of these statements settles Halmos’s third problem for bounded invertible operators. Recent work still treats that problem through additional spectral hypotheses; see [3].

References

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